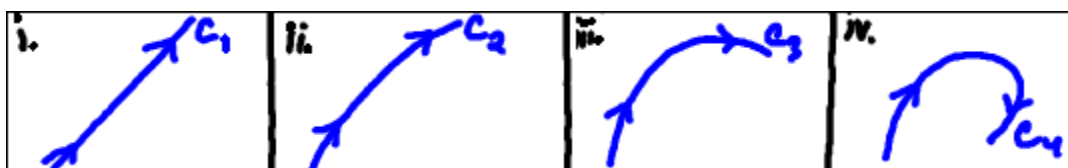


2.5 Curvature

We will now explore the notion of curvature. We begin by reviewing some terminology.

Definition: A parameterization $\mathbf{r}(t)$ is called smooth on an interval I if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$. A curve is smooth if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

Remark: For a smooth curve given by $\mathbf{r}(t)$ when the unit tangent vector, $\mathbf{T}(t)$, turns it does so continuously. When a smooth curve has sharp turns $\mathbf{T}(t)$ will turn quickly. When a curve turns quite slowly, $\mathbf{T}(t)$ will not turn quickly. Consider the pictures below for examples of this idea.



This leads to the following definition.

Definition: The curvature of a smooth curve is $\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$.

Theorem: The curvature of a smooth curve given by $\mathbf{r}(t)$ can be expressed as

$$1) \quad \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}.$$

OR

$$2) \quad \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

Proof:

(1) First, note that by the chain rule that $\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds} \cdot \frac{ds}{dt}$. By the Fundamental Theorem of Calculus we know

that $\frac{ds}{dt} = \|\mathbf{r}'(t)\|$. Since the curve is smooth we know that $\|\mathbf{r}'(t)\| \neq 0$. This means

$$\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds} \cdot \frac{ds}{dt} \Rightarrow \frac{d\mathbf{T}}{ds} = \frac{d\mathbf{T}/dt}{ds/dt} \Rightarrow \frac{d\mathbf{T}}{ds} = \frac{\mathbf{T}'(t)}{\|\mathbf{r}'(t)\|}. \text{ Finally, we have that } \left\| \frac{d\mathbf{T}}{ds} \right\| = \left\| \frac{\mathbf{T}'(t)}{\|\mathbf{r}'(t)\|} \right\| = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$$

which means that $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$ as desired.

(2) We know, by definition, that

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} \Rightarrow \mathbf{r}'(t) = \|\mathbf{r}'(t)\| \mathbf{T}(t) \Rightarrow \mathbf{r}''(t) = \|\mathbf{r}'(t)\|' \mathbf{T}(t) + \|\mathbf{r}'(t)\| \mathbf{T}'(t).$$

We also know, by definition, that $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} \Rightarrow \mathbf{T}'(t) = \|\mathbf{T}'(t)\| \mathbf{N}(t)$ and by (a) we know that

$$\kappa = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} \Rightarrow \|\mathbf{T}'(t)\| = \kappa \|\mathbf{r}'(t)\| \text{ therefore we can see that } \mathbf{T}'(t) = \kappa \|\mathbf{r}'(t)\| \mathbf{N}(t).$$

If we use this to substitute into $\mathbf{r}''(t) = \|\mathbf{r}'(t)\| \mathbf{T}'(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t)$ we obtain that

$$\mathbf{r}''(t) = \|\mathbf{r}'(t)\| \kappa \|\mathbf{r}'(t)\| \mathbf{N}(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t) = \kappa \|\mathbf{r}'(t)\|^2 \mathbf{N}(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t).$$

If we now compute $\mathbf{r}'(t) \times \mathbf{r}''(t)$ we obtain:

$$\begin{aligned} \mathbf{r}'(t) \times \mathbf{r}''(t) &= [\|\mathbf{r}'(t)\| \mathbf{T}(t)] \times [\kappa \|\mathbf{r}'(t)\|^2 \mathbf{N}(t) + \|\mathbf{r}'(t)\|' \mathbf{T}(t)] \\ &= \kappa \|\mathbf{r}'(t)\|^3 [\mathbf{T}(t) \times \mathbf{N}(t)] + \|\mathbf{r}'(t)\| \|\mathbf{r}'(t)\|' [\mathbf{T}(t) \times \mathbf{T}(t)] = \kappa \|\mathbf{r}'(t)\|^3 \mathbf{B}(t) \end{aligned}$$

However, since $\mathbf{B}(t)$ is a unit vector it follows that $\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \kappa \|\mathbf{r}'(t)\|^3$.

$$\text{Therefore, we have that } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

■

Note: Part (b) of this theorem will tend to be more useful since it only requires that we know the vector-valued function $\mathbf{r}(t)$, we don't need to compute the tangent vector $\mathbf{T}(t)$, and we can easily change the parameter if needed.

Example 1: Consider the circle given by $\mathbf{r}(t) = \langle a \cos(t), a \sin(t) \rangle$ where $0 \leq t < 2\pi$ and $a > 0$. Find the curvature of this circle.

We note that $\mathbf{r}'(t) = \langle -a \sin(t), a \cos(t) \rangle$ and $\mathbf{r}''(t) = \langle -a \cos(t), -a \sin(t) \rangle$. So we calculate that:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -a \sin(t) & a \cos(t) & 0 \\ -a \cos(t) & -a \sin(t) & 0 \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + a^2\mathbf{k} \text{ and}$$

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{0+0+a^4} = a^2$$

$$\|\mathbf{r}'(t)\|^3 = \left(\sqrt{a^2 \sin^2(t) + a^2 \cos^2(t)} \right)^3 = \left(\sqrt{a^2} \right)^3 = a^3$$

$$\text{Therefore, } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{a^2}{a^3} = \frac{1}{a}$$

OR (using (1) from the theorem to calculate the curvature)

We note that $\mathbf{r}'(t) = \langle -a \sin(t), a \cos(t) \rangle$. This means

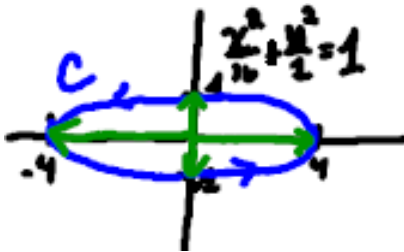
$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{1}{\sqrt{a^2 \sin^2(t) + a^2 \cos^2(t)}} \langle -a \sin(t), a \cos(t) \rangle = \frac{1}{a} \langle -a \sin(t), a \cos(t) \rangle = \langle -\sin(t), \cos(t) \rangle. \text{ We}$$

note from the above calculation that $\|\mathbf{r}'(t)\| = a$ and $\mathbf{T}'(t) = \langle -\cos(t), -\sin(t) \rangle$. So, by the above theorem the

$$\text{curvature is } \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\|\langle -\cos(t), -\sin(t) \rangle\|}{a} = \frac{1}{a}.$$

Note: Intuitively this means that if a circle has a small radius, its curvature is high. This makes sense because small circles turn more abruptly than large circles.

Example 2: Let C be the ellipse $x^2 + 16y^2 = 16$ oriented counterclockwise. Measure the curvature at each endpoint of its major and minor axes.



We note that $\mathbf{r}(t) = 4 \cos(t) \mathbf{i} + \sin(t) \mathbf{j}$.

So, $\mathbf{r}'(t) = -4 \sin(t) \mathbf{i} + \cos(t) \mathbf{j}$ and $\mathbf{r}''(t) = -4 \cos(t) \mathbf{i} - \sin(t) \mathbf{j}$

The endpoints on the positive x and y -axis occur at $t = 0$ and $t = \frac{\pi}{2}$. The curvature at the points on the x -axis are the same and the curvature on the y -axis are the same.

At $t = 0$:

$$\mathbf{r}'(0) = 1\mathbf{j} \Rightarrow \|\mathbf{r}'(0)\|^3 = 1$$

$$\mathbf{r}''(0) = -4\mathbf{i}$$

$$\mathbf{r}'(0) \times \mathbf{r}''(0) = 4\mathbf{k} \Rightarrow \|\mathbf{r}'(0) \times \mathbf{r}''(0)\| = 4$$

$$\text{Therefore, } \kappa(0) = \frac{\|\mathbf{r}'(0) \times \mathbf{r}''(0)\|}{\|\mathbf{r}'(0)\|^3} = \frac{4}{1} = 4$$

At $t = \frac{\pi}{2}$:

$$\mathbf{r}'\left(\frac{\pi}{2}\right) = -4\mathbf{i} \Rightarrow \left\|\mathbf{r}'\left(\frac{\pi}{2}\right)\right\|^3 = 4^3 = 64$$

$$\mathbf{r}''\left(\frac{\pi}{2}\right) = -\mathbf{j}$$

$$\mathbf{r}'\left(\frac{\pi}{2}\right) \times \mathbf{r}''\left(\frac{\pi}{2}\right) = 4\mathbf{k} \Rightarrow \left\|\mathbf{r}'\left(\frac{\pi}{2}\right) \times \mathbf{r}''\left(\frac{\pi}{2}\right)\right\| = 4$$

$$\text{Therefore, } \kappa\left(\frac{\pi}{2}\right) = \frac{\left\|\mathbf{r}'\left(\frac{\pi}{2}\right) \times \mathbf{r}''\left(\frac{\pi}{2}\right)\right\|}{\left\|\mathbf{r}'\left(\frac{\pi}{2}\right)\right\|^3} = \frac{4}{64} = \frac{1}{16}$$